

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
2	(a)		$I = 0.0012 \text{ A}$ and $R_{\text{tot}} = 5000 \Omega$ or $V_{\text{therm}} = 4.80 \text{ V}$ or $\frac{R_{\text{th}}}{1000 \Omega} = \frac{4.8 \text{ V}}{1.2 \text{ V}}$ or any other correct and relevant potential divider equation or by implication (1) $R_{\text{therm}} = 4000 \Omega$ (1) $\theta = 15^\circ$ ecf on R_{therm} (1)		3		3	3	
	(b)		Graph gradient changes [with θ] or graph not straight [or voltmeter reading doesn't vary linearly with R_{therm}] or ΔV not proportional to ΔR (1) n not constant [with varying θ] / he is wrong [only award if backed up by reasonable argument] (1)			2	2	1	
	(c)		So <u>thermistor</u> [accept: 'it'] doesn't heat up (1) Reference to electrical heating [even of circuit or resistor] or masking response to surroundings (or equiv) (1)		2		2		
			Question 2 total	0	5	2	7	4	0